MIL-E-22345(Wep)
11 October 1960

MILITARY SPECIFICATION

EXTINGUISHER, FIRE, DRY CHEMICAL,
WHEELED, 150-POUND CAPACITY

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope: The fire extinguisher covered by this specification shall be designed for fighting fires in and around aircraft on the ground. Such fires include tire and wheel fires or fires which might occur during defueling or refueling activities. The equipment described shall be designed to permit towing at moderate speeds over uneven terrain behind military vehicles.

1.2 Classification: The fire extinguisher covered by this specification shall be furnished in the size shown in Figure 1, charged as specified.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications of issue on the date of invitation for bids form a part of this specification. Where there is a conflict between this specification and the referenced specifications, drawings, standards or other publications this specification shall govern:

SPECIFICATIONS:

FEDERAL

UU-P-271	-	Paper, wrapping, waterproofed, Kraft
PPP-B-601	-	Boxes, wood, cleated, plywood
PPP-B-621	-	Boxes, wood, nailed and lock-corner
NN-P-515	-	Plywood, container grade

MILITARY

MIL-A-140	-	Adhesive, water-resistant, waterproof, barrier-material
MIL-R-13689	-	Reflectorized sheeting, adhesive(Retro-reflective)
JAN-P-108	-	Packaging and packing for overseas shipment, boxes, fiberboard(V-board and W-board), exterior and interior
MIL-F-19563	-	Fire extinguishing agent, dry chemical, foam compatible
MIL-F-22287	-	Fire extinguishing dry chemical, potassium bicarbonate type

FSC-4210

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STANDARDS:

MILITARY

MIL-STD-129 - Marking for shipment and storage

PUBLICATIONS:

AIR FORCE - NAVY AERONAUTICAL BULLETINS

No. 143 - Specifications and Standards; use of

(When requesting specifications, standards and publications refer to both the Title and Number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, Attention, Code CDS)

2.2 **Other Publications**: The following publications, of the issue in effect on date of bids, unless otherwise stated, form a part of this specification:

NAVAL RESEARCH LABORATORY

NRL Report 5183

(Copies of Naval Research Laboratory Reports may be obtained upon application to the Director, Naval Research Laboratory, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Structural Dimensions: The structural dimensions shall be as indicated in Figure 1.

3.1.1 Specifications and Standards: Specifications and standards for all materials, parts, and government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraphs.

3.2 General: The fire extinguisher described in this specification shall be complete and shall include a funnel for use in charging the unit with dry chemical. Using hand maintenance tools, the extinguisher shall be capable of being charged with dry chemical and otherwise made ready for operation in a few minutes' time. The fire extinguisher shall be constructed to meet all the performance requirements and standards when tested as specified herein.

3.2.1 Preproduction Sample: In all respects not specifically covered by

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this specification, the fire extinguisher shall be equal to the preproduction sample. The structural and operational characteristics of this sample shall be evaluated also for acceptability from a standpoint of safety and efficiency in operation and for conformance with generally accepted construction practice. Unless otherwise specified preproduction samples as indicated in paragraph 4 shall be submitted for examination, inspection and test to determine compliance with all provisions of this specification. The first fire extinguishers produced under the terms of the contract, following notice of award, may be submitted for approval as production samples and approval shall be obtained prior to fabricating subsequent items. Fabrication of additional items prior to approval of the production samples shall be at the contractor's risk, and such additional items are subject to rejection in the event they fail to meet the requirements of this specification. Preproduction sample fire extinguishers, when approved will be used as a measurement guide for all fire extinguishers subsequently manufactured in a given contract under this specification and will be presented for inspection as the last units for delivery under the contract. When specified examinations and testing of the preproduction samples shall be conducted by the Naval Research Laboratory, Washington, D. C. The production units shall when practicable be examined and tested at the manufacturer's plant by a representative of the contracting office.

3.3 Materials: Materials used in the construction of these wheeled dry chemical fire extinguishers shall be of the best quality and workmanship, entirely suitable for the purpose intended, with the strength and materials required for safety, reliability and resistance to corrosion.

3.4 Design and Construction: Fire extinguishers shall be designed and constructed to permit easy operation, recharging, inspection and maintenance. The extinguisher shall consist of the following:

Dry Chemical Tank	Nozzle
Nitrogen Cylinder	Wheels and Axles
Pressure Regulator	Hose Rack and Hose Confiner
Discharge Hose	Handle with Towing Loop

3.4.1 Dry Chemical Tank: The dry chemical tank shall be of welded steel construction with a minimum inside diameter fill opening of 4 inches. Each dry chemical tank shall be hydrostatically tested at 600 psi and shall have a minimum rupture strength of not less than five times its normal working pressure.

3.4.1.2 Dry chemical tank shall be provided with a gas tube which employs a check valve to prevent syphoning of dry chemical back into the gas tube and pressure regulator.

3.4.1.3 Fill Cap: Fill cap on dry chemical tank shall be provided with safety vent holes so that cap is pressure venting while at least 3-1/2

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threads are still engaged. The fill cap arrangement shall be such that the male threads are a part of the tank and the female threads are a part of the cap.

3.4.2 Nitrogen Cylinder: Nitrogen cylinder shall be constructed and marked in accordance with ICC-3A-2015. 110 cubic feet capacity at 2215 psi.

3.4.2.1 Nitrogen Cylinder Valve: Nitrogen cylinder valve shall be constructed in accordance with Compressor Gas Association (CGA) standards for cylinder valve outlet connection for water pumped nitrogen. A 3000 psi pressure gage indicating cylinder pressure with the valve in either the open or shut position shall be provided.

3.4.3 Pressure Regulator: Nitrogen pressure regulator provided shall be mounted adjacent to the dry chemical tank in a position where it is protected from mechanical injury. The regulator shall be set by the manufacturer to the desired working pressure and the setting screw secured in that position by a suitable sealing mechanism. Fittings between the discharge side of the regulator and the dry chemical tank shall have a rupture pressure of not less than 1500 psi. Connections between the gas cylinder and the regulator shall have a rupture pressure of not less than 5000 psi. Gas flow capacity of the regulator shall insure the full working pressure in the dry chemical tank within 20 seconds of opening of the cylinder valve. Regulator shall be provided with a spring loaded relief valve set to discharge at 300 psi and handle the full flow of the regulator.

3.4.4 Discharge Hose: The dry chemical discharge hose shall be a 50-foot length of 3/4 inch, Underwriters' Laboratories, Inc. labeled chemical engine hose. The hose and couplings shall be capable of withstanding a hydrostatic pressure of 600 psi for a period of 5 minutes.

3.4.4.1 Swivel Coupling: A swivel coupling shall be provided for connection of the 3/4 inch dry chemical hose to the dry chemical outlet.

3.4.5 Nozzle: The dry chemical discharge nozzle shall be made of cast aluminum alloy with integral component parts of aluminum and stainless steel.

3.4.5.1 Nozzle construction shall permit movement of control handle to three positions ... off, fan stream and straight stream. Nozzle shall be pressure tested after assembly at 220 to 240 psi. Allowable leakage shall not exceed the formation of slow bubbles in soap and water test.

3.4.6 Wheels and Axles: Wheels shall be mounted on steel axles located near the center of balance of a fully charged extinguisher.

3.4.6.1 Rubber Tires: Wheels shall be equipped with solid rubber tires molded to the wheel.

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3.4.7 Hose Rack and Hose Confiner: A hose rack shall be provided which allows coiling of the hose around an upper and lower hose support. A hose confiner shall be provided to contain the hose laterally so hose does not interfere with free rotation of wheels.

3.4.7.1 Nozzle Confiner: A nozzle holder shall be provided including a cadmium plated steel spring to secure the nozzle in its holder.

3.4.8 Handling with Towing Loop: The extinguisher handle shall be equipped with a steel towing loop securely welded to the handle thereby permitting quick attachment to pintles on military vehicles and subsequent towing.

3.4.9 Attached to the extinguisher shall be a metal plate specifying the safe recommended maximum towing speed.

3.4.10 Instruction Manual: Two copies of the manufacturer's standard instruction manual shall be furnished and packed with each extinguisher.

3.5 Weight: The weight of a fully charged extinguisher shall not exceed 425 pounds.

3.6 Capacity: Dry chemical capacity under normal filling conditions shall be a minimum of 140 pounds without rodding or packing.

3.7 Performance: Fire extinguishers shall satisfy performance requirements as specified in Section 4.

3.8 Instruction Plate: An etched aluminum name plate shall be permanently attached to the side of the dry chemical tank. The marking on the plate shall include:

1. Name and model number of the unit.

2. Name and address of the manufacturer.

3. Complete, clear and concise directions for proper operation of the unit in red letters large enough to be easily read.

4. "After using" and maintenance instructions shall be in smaller, black letters at the bottom of the plate.

5. A statement that the gas cylinder be replaced if the pressure drops below a specified point.

3.9 Identification of Product

3.9.1 Name Plate: The name plate shall permanently and legibly show that the extinguisher is of the dry chemical type.

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3.10 Workmanship: Workmanship shall be in accordance with high grade commercial practice covering this class of equipment.

3.11 Color Identification: All exposed steel surfaces, except thread, gasket, and axle contact points shall be thoroughly cleaned of rust, scale and loose weld and primed with zinc chromate primer and painted with a red enamel paint.

3.11.1 A band of reflectorized tape six inches wide and of a color specified by the Contracting Officer shall be attached around the top of the dry chemical tank above the name plate.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor Inspection: Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission of the equipment for official inspection and acceptance. The contractor shall utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and test shall be kept complete and available to the Government as specified in the contract order.

4.2 Notification: The contractor shall notify the Government Inspector and the Chief, Bureau of Naval Weapons (Code SEEQ) in sufficient time prior to presenting the preproduction sample for acceptance tests so that government representatives may witness all tests. All test equipment necessary for conducting tests in accordance with this specification shall be supplied by the contractor. It is to be understood that the methods described herein will in general be used.

4.3 Preproduction Sample: The preproduction sample is defined as a production sample of the product which the manufacturer will subsequently produce in fulfillment of the contract. As soon as practicable after the award of the contract or order and prior to submission of any complete unit for final acceptance, the contractor shall furnish a preproduction sample for examination and test to determine conformance of this specification. Approval of the preproduction sample by the activity concerned shall not relieve the contractor of his obligation to supply equipment conforming to this specification. Any changes to or deviation from the preproduction sample shall be subject to approval of the contracting officer.

4.4 Inspection and Tests: The equipment and preparation for delivery thereof shall be given such inspection and tests as may be necessary to determine conformance with this Specification. The Government reserves the right to use such additional information or methods deemed necessary to determine compliance with the requirements of this specification.

4.5 Preproduction Test Sample: Unless otherwise specified in the contract or order, tests of preproduction test sample shall be made at the place of manufacture.

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4.5.1 Preproduction Sample: The preproduction sample shall be examined, inspected and tested to determine compliance with all provisions of paragraph 4 of this specification. This sample shall be examined and, tested also to determine acceptability on the basis of conformance to accepted engineering design practice and of efficiency and safety of the operation characteristics normally available in similar commercial items.

4.6 Classification of Tests

4.6.1 Preproduction Tests: Preproduction tests are those tests accomplished on sample representative of the preproduction fire extinguishing units to determine that it meets the requirements of this specification.

4.6.2 Acceptance Tests: Acceptance tests are those tests accomplished on fire extinguishing units submitted for acceptance under contract.

4.7 Preproduction Samples

4.7.1 Preproduction Tests: The preproduction fire extinguishers shall be subjected to the applicable tests specified in 4.

4.7.2 Acceptance Test: Acceptance test shall be as follows:

a. Each unit manufactured under a given contract shall be given the inspection and tests outlined in 4.8 through 4.8.1.7.

b. The Government Inspector shall test not less than one unit in over 50 units produced under a given contract as outlined in 4.

4.7.3 Disqualification: Failure of any qualification test specimen to pass any or all specified tests shall be cause for refusal of the government to conduct additional tests until it has been proven, in writing, to the satisfaction of the procuring agency that the faults revealed by the tests have been corrected.

4.8 Inspection and Tests

4.8.1 Visual, Dimensional and Primary Function Inspection: The unit shall be subjected to the following examinations.

4.8.1.1 General Inspection: Examine the name plate for conformance to paragraphs 3.8 and 3.9.1. Inspect and measure the marking band for conformance to paragraph 3.11. Examine welds for smoothness and proper beads. Examine and measure container for conformance to Figure 1. Weigh the unit to determine compliance with paragraph 3.5.

4.8.1.2 Dry Chemical Tank: Examine interior of dry chemical tank to make certain gas tube is equipped with a check valve to conform with requirements of paragraph 3.4.1.2.

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4.8.1.3 Fill Cap: Examine fill cap to assure it complies with requirement of paragraph 3.4.1.3.

4.8.1.4 Nitrogen Cylinder: Examine nitrogen cylinder to see it complies and is marked in accordance with the requirements of paragraph 3.4.2.

4.8.1.5 Pressure Regulator: Pressure regulator shall be tested prior to assembly on the extinguisher to make certain the low pressure setting lies between 220 and 240 psi.

4.8.1.6 Nozzle: The nozzle shall be examined to make certain it complies with requirements of paragraph 3.4.5.

4.8.1.7 Rubber Tires: Rubber tires shall be examined to make certain they are securely molded to the wheels, not cemented to the wheels.

4.8.1.8 Discharge Tests: The effective discharge time shall be not less than 60 seconds and a minimum of 80 per cent of the capacity shall be discharged within this time. The time is taken as being that between the time the first dry chemical leaves the nozzle and the first sputtering or noticeable thinning of the dry chemical discharge stream takes place.

The discharge tests shall be conducted with dry chemical extinguishing agents conforming with specifications MIL-F-19563 and MIL-F-22287 to demonstrate satisfactory operation with these materials.

4.8.1.9 Intermittent Operation Test: The fully charged and pressurized extinguisher shall be discharged in 5-second on and 5-second off periods. The discharge shall be free at all times and at least 80 per cent of the total capacity shall be discharged.

4.8.1.10 Abuse Tests: The extinguisher unit shall be capable of withstanding the following abuse without impairing its proper operation:

1. A free ~~pull~~ FA 44 of the nozzle attached to the hose from a height of 3 feet to a concrete floor.

2. Pulling the unit from a 1-foot high platform onto a concrete floor.

3. One wheel striking a 6-inch high concrete curb while being towed at a moderate speed.

5. PREPARATION FOR DELIVERY

5.1 Application: The packaging, packing and marking requirements specified herein apply only to direct purchases by or direct shipments to the government.

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5.2 Packaging

5.2.1 Unit Packaging: Unless otherwise specified, unit packages shall consist of the empty uncrated extinguisher plus the dry chemical charge contained in three 50-pound capacity metal containers.

5.2.1.1 Cushioning: Unless otherwise specified, adequate cushioning made of corrugated fiberboard shall be provided to protect the name plates. A plastic thread protector shall be employed to protect exposed, uncoupled male threaded connections. The nitrogen cylinder valve shall be protected by means of a steel cylinder cap threaded to the collar of the nitrogen cylinder on each extinguisher.

5.2.2 Packing: Unless otherwise specified, all items shall receive domestic packing.

5.2.2.1 Domestic Packing: Unless otherwise specified, the wheeled extinguisher shall not be charged with dry chemical and shall not be crated. The dry chemical charge shall be shipped in steel containers to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery. Dry chemical containers and extinguishers shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.2.2.2 Overseas Packing: Unless otherwise specified, the wheeled extinguisher, complete with its dry chemical charge, shall be shipped in a wire bound crate such as is used in shipping similar commercial units via land and water freight to overseas destinations.

5.3 Marking and Labeling

5.3.1 Unit Packages: Each unit package, including both extinguisher and steel containers for dry chemical shall be durably and legibly marked with the following information in such a manner that markings will not become damaged when packages are opened:

EXTINGUISHER, DRY CHEMICAL, WHEELED

AN Part No.
Stock No. (USAF or NAVY, as applicable)
Name of Manufacturer (If not same as contractor)
Name of Contractor
Order or Contract No.

6. NOTES

6.1 Ordering Data: Requisitions, contracts and orders shall state the capacity of fire extinguisher and whether overseas packing is desired.

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6.2 It is to be understood that extinguishers supplied under contract shall be identical in every respect to the sample tested and found satisfactory (and to the detail drawings thereof), except for changes previously approved by the Government. Any unapproved changes from the qualification sample (or from the drawings) shall constitute cause for rejection.

6.3 The extinguishing agent used with this extinguisher may be dry chemical potassium bicarbonate type conforming to Specification MIL-F-22287(Wep).

6.4 Intended Use: The fire extinguisher covered by this specification shall be designed for fighting fires in and around aircraft on the ground. Such fires include tire and wheel fires or fires which might occur during defueling or refueling activities. The equipment described shall be designed to permit towing at moderate speeds over uneven terrain behind military vehicles.

6.5 Information to Accompany Bid: Bidder should submit a positive statement whether or not item bid upon fully conforms to the terms of this specification. In addition, bidder should furnish complete descriptive literature, drawings or photographs, and complete technical data covering the equipment he proposes to furnish, identifying the equipment by the name of the manufacturer and the model number. Lack of specific and complete information will be sufficient cause for rejection of bid(s). Where the bidder's product differs from the specific requirements, each point of difference should be clearly described. This requirement is set forth to facilitate the review of bids and is not to be construed by the bidder as waiving any of the requirements of the specification.

6.6 Guaranty: Invitation for bids and contracts or orders should specify the following guaranty clause, except in those instances where it is determined that the inclusion of the guaranty clause would not be in the best interest of the Government:

a. The contractor guarantees that at the time of delivery thereof the articles provided for under this contract will be free from any defects in material or faulty workmanship or improper design and will conform to the requirements of this contract. If the Government discovers any such defects or nonconformance within a period of one year after delivery, it may require the contractor (1) to correct or replace (as designated by the Contracting Officer) the defective or nonconforming articles or part thereof; or (2) to repay such portion of the contract price as is equitable in the circumstances. Notice of the defect or nonconformance shall be given to the contractor within a reasonable time after delivery thereof.

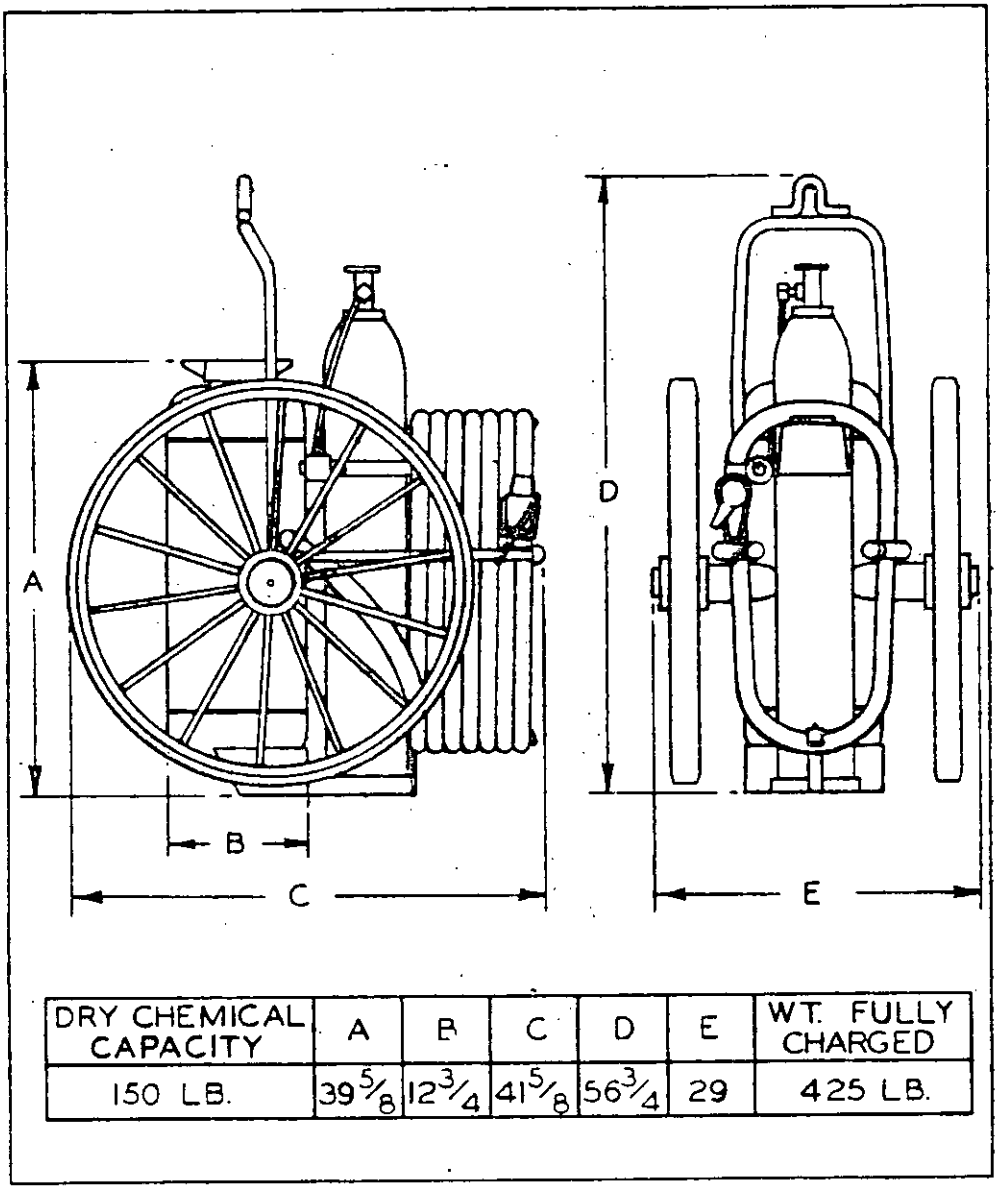
b. If required by the Government to correct or replace the defective or nonconforming article or part thereof, the contractor shall with all possible speed made such correction or replacement. When such correction or replacement requires transportation of the article or part thereof, shipping

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costs, not exceeding usual charges from site of tractor operation within the continental United States, from the nearest designated port of embarkation to the contractor's plant and return, shall be borne by the contractor; the Government shall bear all other shipping costs. This guaranty shall then continue as to corrected or replaced articles, or, if only parts of such articles are corrected or replaced, to such corrected or replaced parts; until one year after date of delivery.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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DRY CHEMICAL CAPACITY	A	B	C	D	E	WT. FULLY CHARGED
150 LB.	$39\frac{5}{8}$	$12\frac{3}{4}$	$41\frac{5}{8}$	$56\frac{3}{4}$	29	425 LB.

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